ArgiSense Sensor IO Mapping

This document records the firmware-facing IO mapping for the two ArgiSense sensor products:

- `argisense_mp_u575rg`: methane + pressure sensor firmware.
- `argisense_ph_u575rg`: pH sensor firmware.

The mapping is based on the current board DTS files and the reviewed `Sensor_Platform_V1 (1).pdf` schematic. Conflicts and open hardware items are listed at the end so PCB routing and firmware changes can be tracked together.

Shared Service IO

Both products keep the service/debug pins consistent where possible.

Function	DTS node/property	Firmware pin mapping	Notes
Debug console	`usart1`	`PA9` TX, `PA10` RX	Default console and shell unless a USB snippet overrides it.
RS485 Modbus/DFU	`usart2`	`PA2` TX, `PA3` RX, `PA1` DE	`de-enable` is enabled in DTS.
RS485 termination	`rs485-termination-gpios`	`PB0`	Runtime setting controls this GPIO.
USB-C device	`otgfs` through USB snippets	`PA12` DP, `PA11` DM	Used for shell/console and MCUmgr USB update profiles.
SWD debug	hardware debug	`PA13` SWDIO, `PA14` SWCLK	Reserved for debugger/programmer.
User LED	`led0`	`PB7`	Board status LED.
User button	`sw0`	`PC13`	GPIO key input.
Flash storage	`storage_partition`	internal flash offset `0x000f0000`	Zephyr Settings/NVS backend.

Methane + Pressure Board

Board target:

```
argisense_mp_u575rg
```

DTS:

```
boards/argisense/argisense_mp_u575rg/argisense_mp_u575rg.dts
```

Methane Sensor

Signal	Schematic net	DTS node/property	Firmware mapping	Status
Dynament UART TX from MCU	`METH_UART_TXD`	`uart4`	`PC10 / UART4_TX`	Firmware-valid mapping.
Dynament UART RX to MCU	`METH_UART_RXD`	`uart4`	`PC11 / UART4_RX`	Firmware-valid mapping.
Methane connector pins	`SENSOR_COMM1..4`	none	connector-side nets	No separate DTS mapping needed when populated as the UART path.
Methane power rail	`+3V3_PRE`	`pre-power-gpios`	`PC8`	Kept on by default for Dynament continuous operation.

Conflict:

The reviewed schematic shows METH_UART_TXD on PC8 and METH_UART_RXD on PC7. The STM32U575RGT6 pinctrl data used by Zephyr does not provide UART TX/RX alternate functions on PC8/PC7. The current firmware therefore keeps the Dynament sensor on UART4 PC10/PC11. The PCB should route the methane UART to PC10/PC11, or another valid UART TX/RX pair, before relying on the DTS on real hardware.

Pressure Sensor

Signal	Schematic net	DTS node/property	Firmware mapping	Status
SPI chip select	PRES_SEN_CSB	spi1 cs-gpios, pressure-cs-gpios	PA4 active-low	Mapped.
SPI clock	PRES_SEN_SCLK	spi1	PA5 / SPI1_SCK	Mapped.
SPI MISO	PRES_SEN_SDO	spi1	PA6 / SPI1_MISO	Mapped.
SPI MOSI	PRES_SEN_SDI/SDA	spi1	PA7 / SPI1_MOSI	Mapped.
Pressure power/shutdown	PRES_SEN_PS	pressure-ps-gpios	PB1	Logical active state is handled in firmware.
Pressure device	MS580305BA01-00	pressure_ms5803	SPI device CS0	Driver implemented in drivers/sensor/ms5803_05ba.

Shared Sensors And Outputs

Function	Schematic net	DTS node/property	Firmware mapping	Status
GP8302 DAC0 for methane	O_D_UC_I2C1_SCL/SDA	current-loop-dac0, i2c1	PB8/PB9, address 0x58	Mapped.
GP8302 DAC1 for pressure	O_D_UC_I2C2_SCL/SDA	current-loop-dac1, i2c2	PB10/PB14, address 0x58	Mapped.
DAC boost enable	O_D_UC_PWR_DAC	dac-power-gpios	PC6	Mapped.
DAC alarm	O_D_DAC_ALARM / O_D_DAC_ALARM1	dac-alarm-gpios	PC7	One alarm GPIO is mapped.
HTU21D humidity/temp	MCU_I2C2_SCL/SDA	humidity-sensor, i2c2	PB10/PB14, address 0x40	Mapped.
AT24C512C EEPROM	MCU_I2C1_SCL/SDA	eeprom0, i2c1	PB8/PB9, address 0x50	Mapped.
Analog rail enable	U_O_D_ANA_PWR_EN	analog-power-gpios	PC9	Mapped.
MCU analog monitor	MCU_ADC3_INP0 equivalent	thermistor-adc, adc1	PC0 / ADC1_IN1	Mapped as a local analog monitor.

Open item:

The schematic exposes a second DAC alarm net (O_D_DAC_ALARM2) for the second GP8302 output path. The methane + pressure DTS currently maps only dac-alarm-gpios. If both GP8302 alarm pins must be monitored independently, add a dac1-alarm-gpios property and update the methane + pressure power or diagnostic code to read it.

pH Board

Board target:

```
argisense_ph_u575rg
```

DTS:

```
boards/argisense/argisense_ph_u575rg/argisense_ph_u575rg.dts
```

pH ADC: ADS124S08

Signal	Schematic net	DTS node/property	Firmware mapping	Status
SPI chip select	`ADC_SPI_CS`	`spi1 cs-gpios`	PA4 active-low	Mapped.
SPI clock	`ADC_SPI_SCLK`	`spi1`	`PA5 / SPI1_SCK`	Mapped.
SPI DOUT from ADC	`ADC_SPI_DOUT`	`spi1`	`PA6 / SPI1_MISO`	Mapped.
SPI DIN to ADC	`ADC_SPI_DIN`	`spi1`	`PA7 / SPI1_MOSI`	Mapped.
ADC reset	`ADC_RESET`	`reset-gpios`	PB12 active-low	Mapped.
ADC start/sync	`ADC_START`	`start-sync-gpios`	PB13 active-high	Mapped.
ADC data ready	`ADC_SRDY`	`drdy-gpios`	PB15 active-low	Mapped.
PT1000 input	`PT1000_P/N`	ADS124S08 inputs	`AIN2/AIN3`	Mapped in ADC channel metadata.

pH Bias DAC: AD5675

Signal	Schematic net	DTS node/property	Firmware mapping	Status
DAC I2C clock	`DAC_I2C_SCL`	`i2c3`	`PC0`	Mapped.
DAC I2C data	`DAC_I2C_SDA`	`i2c3`	`PC1`	Mapped.
DAC reset	`DAC_RESET`	`reset-gpios`	PC2 active-low	Mapped.
DAC LDAC	`LDAC*`	optional `ldac-gpios`	not routed to MCU on pH_V1_0	Hardware option, not a DTS pin.
DAC reference	REF3025 2.5 V	`reference-millivolt`	`2500`	Metadata used for output scaling.

Conflict / hardware note:

`LDAC*` is tied low through `R73 0R` and exposed at `TP17` on the reviewed pH schematic. The AD5675 driver supports optional `ldac-gpios`, but the current board DTS intentionally omits it because there is no MCU route. If a later PCB revision needs MCU-controlled simultaneous DAC updates, route `LDAC*` to a free GPIO and add `ldac-gpios = <... GPIO_ACTIVE_LOW>;` to the AD5675 node.

pH Analog Front-End And Outputs

Function	Schematic net	DTS node/property	Firmware mapping	Status
ADC clean 3V3 enable	`U_O_D_3V3_EN`	`adc-power-gpios`	`PC3`	Mapped.
Electrode zero switch	`ELECTRODE_ZERO_CTRL`	`electrode-zero-gpios`	`PC4`	Mapped.
Gate/analog monitor	`GATE_BUFF_ADC` / monitor input	`mcu-adc`, `io-channels`	`PC5 / ADC1_IN14`	Mapped.
DAC boost enable	`O_D_UC_PWR_DAC`	`dac-power-gpios`	`PC6`	Mapped.
GP8302 DAC0 alarm	`O_D_DAC_ALARM1`	`dac-alarm-gpios`	`PC7`	Mapped.
Sensor/pre rail enable	+3V3_PRE control	`pre-power-gpios`	`PC8`	Mapped.
Analog rail enable	`U_O_D_ANA_PWR_EN`	`analog-power-gpios`	`PC9`	Mapped.
GP8302 DAC1 alarm	`O_D_DAC_ALARM2`	`dac1-alarm-gpios`	`PB2`	Mapped.
GP8302 DAC0 output	`O_D_UC_I2C1_SCL/SDA`	`current-loop-dac0`, `i2c1`	`PB8/PB9`, address `0x58`	Outputs pH.
GP8302 DAC1 output	`O_D_UC_I2C2_SCL/SDA`	`current-loop-dac1`, `i2c2`	`PB10/PB14`, address `0x58`	Outputs PT1000 temperature.
HTU21D humidity/temp	n/a on pH_V1_0	no pH DTS node	n/a	Not used by the pH product; temperature comes from PT1000 through ADS124S08.
AT24C512C EEPROM	`MCU_I2C1_SCL/SDA`	`eeprom0`, `i2c1`	`PB8/PB9`, address `0x50`	Mapped.
LT3582 analog rail	analog rail I2C block	`ph_analog_rail`	`i2c2`, placeholder `0x2c`	Node disabled until address/programming is verified.

Conflict Summary

Area	Affected board	Conflict / open item	Firmware action	Hardware action
Dynament methane UART	`argisense_mp_u575rg`	Schematic shows `METH_UART_TXD/RXD` on `PC8/PC7`, but those pins are not valid UART TX/RX pins in Zephyr STM32U575RGT6 pinctrl.	Keep firmware on `UART4` `PC10/PC11` until PCB route is corrected.	Route methane UART to `PC10/PC11` or another valid UART pair.
PC7/PC8 reuse	`argisense_mp_u575rg`	Current DTS uses `PC7` for DAC alarm and `PC8` for pre-power, while the reviewed schematic labels `PC7/PC8` as methane UART.	Do not move UART to `PC7/PC8`; document as routing conflict.	Resolve schematic pin allocation before PCB release.
AD5675 LDAC	`argisense_ph_u575rg`	`LDAC*` is not routed to MCU; it is tied low through `R73 0R` and exposed at `TP17`.	Driver supports optional `ldac-gpios`; DTS omits it now.	Route `LDAC*` to MCU only if firmware-controlle d LDAC is required.
Second GP8302 alarm	`argisense_mp_u575rg`	Schematic exposes `O_D_DAC_ALARM2`; MP DTS maps only one `dac-alarm-gpios`.	Add `dac1-alarm-gpios` and diagnostics if independent alarm monitoring is required.	Confirm whether both alarm outputs are populated and need MCU monitoring.
LT3582 pH rail programming	`argisense_ph_u575rg`	I2C address/programming requirement is not verified.	Keep `ph_analog_rail` disabled and control rail by enable GPIO.	Verify LT3582 address and startup programming on hardware.

Recommended Next Checks

- Decide final methane UART pins in the schematic and update the PCB route before hardware bring-up.
- Decide whether methane + pressure needs independent monitoring of both GP8302 alarm pins.
- Decide whether pH needs MCU-controlled AD5675 `LDAC*`; if yes, route the pin in the next PCB revision.
- Validate the pH LT3582 rail IC address and whether it needs runtime I2C programming.
- After schematic updates, rebuild both targets and compare generated `zephyr.dts` against the final netlist.